BLKEXPORT User Guide

June 2026
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Software Version 1.1

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2. Quick Start: DXF to BOM in 3 steps

1. **Open**
   File > Open DXF or Ctrl+O. Pick your drawing.
   → Check *Integrity Summary* shows PASS.

2. **Review**
   Click the tabs: **Attributed** | **Non-Attributed** | **Anonymous**
   → These are your blocks, auto-sorted.

3. **Export**
   Ctrl+E = Full block schedule CSV (purple button)
   Ctrl+B = Summarised BOM CSV (green button)

   Done. Open the CSV in Excel.

3. Overview

BLKEXPORT is a desktop utility designed to process DXF drawings and identify block insertions. The application categorises blocks into Anonymous Blocks, Non-Attributed Blocks, and Attributed Blocks, then allows the results to be exported to CSV files. It also provides a summarised BOM (Bill of Materials) export and full block schedule export.

4.Who is BLKEXPORT for?

BLKEXPORT is designed for anyone who needs to generate a Bill of Materials (BOM) from AutoCAD drawings. It automatically extracts all blocks from a drawing and compiles them into a BOM.

Landscape architects can use BLKEXPORT to count every plant and tree in a landscape design. M&E engineers can generate a list of all electrical components in a building. Process engineers can produce a BOM from P&ID drawings to schedule valves, vessels, pumps, and other equipment.

If your drawings use blocks, BLKEXPORT can generate your BOM automatically.

5. System Requirements & Known Limits

Minimum Requirements

- Operating System: Windows 11 Pro
- Installed RAM 8 GB
- Display 1366×768 resolution
- Storage 50 MB free space
- System type 64-bit operating system, x64-based processor

Tested Limits

These limits reflect the test environment used for v1.1 validation. BLKEXPORT does not impose its own block count limit, but was tested with files created in progeCAD 2021.

Block count: Tested up to 100,000 blocks per DXF file. This upper bound is imposed by progeCAD 2021, which was used to generate the test files. BLKEXPORT successfully processed all files up to this limit. Behaviour above 100,000 blocks is untested.

DXF file size: Successfully tested up to 46 KB for ASCII DXF and 29 KB for Binary DXF containing 100,000 blocks.

Note: File size depends on geometry and attribute complexity. The 46 KB test file used simple blocks with minimal geometry. Drawings with complex blocks or extensive attribute data will be larger and may load more slowly.

Memory usage: Loading a 100,000-block DXF file used approximately 1.8 GB of RAM on the reference test machine. Systems with 4 GB RAM may experience slowdowns or paging to disk when processing files near this limit.

Attribute length: Maximum 255 characters per individual attribute, as defined by the DXF specification. Total attributes per block are limited by available memory.

Supported DXF formats: AutoCAD 2000 through AutoCAD 2018, inclusive. Both ASCII and Binary DXF variants are supported.

Unsupported formats: DWG format is not supported. Convert to DXF first using AutoCAD or progeCAD.

Unsupported block types: Dynamic Blocks and Smart Blocks are not supported. They will be ignored during processing and excluded from counts/exports.

DXF load performance: Loading a DXF file containing 100,000 blocks takes approximately **2.5 minutes** on the reference test machine. During loading, BLKEXPORT may appear unresponsive while the file is being processed. Actual load times depend on host machine performance, DXF complexity, and whether the file is on a local SSD or network drive. Once the DXF file is loaded, exporting the summarised BOM and full block schedule to CSV is instantaneous.

Reference Test Machine Specification

- Operating System: Windows 11 Pro
- Operating System Version: 25H2
- Processor 13th Gen Intel(R) Core(TM) i9-13900 (2.00 GHz)
- Installed RAM 32.0 GB (31.6 GB usable)
- Graphics card Intel(R) UHD Graphics 770 (128 MB)
- Storage 1TB SSD
- Display 2560 × 1440 resolution
- System type 64-bit operating system, x64-based processor

Notes

- If *Integrity Summary* shows FAIL, check file format first.
- Network drives supported, but local SSD recommended for files >10MB.

6. Supported Files

Supported formats

- AutoCAD 2000 through AutoCAD 2018, inclusive. Both ASCII and Binary DXF variants are supported.

- DWG not supported.

7. Install BLKEXPORT Software

The BLKEXPORT software comes with a built-in installer.

1. **Run the installer**
   Right-click setup.exe and choose **Run as administrator**.
2. **Choose install location**
   Keep the default path C:\Program Files\BLKEXPORT\ and click **Next**.
3. **Set user access**
   Select **Install for all users**, then click **Install**.
4. **Finish**
   When installation completes, the BLKEXPORT desktop icon will be created.

   BLKEXPORT is now installed. Double-click the desktop icon to launch the application.

8. Uninstall BLKEXPORT Software

1. **Launch Control Panel**
   Press Win + R, type control, and press Enter.
2. **Open Programs and Features**
   Click **Programs > Programs and Features**.
3. **Uninstall BLKEXPORT**
   Scroll to **BLKEXPORT**, right-click, and choose **Uninstall**. Click **Yes** to confirm.
4. **Done**
   BLKEXPORT is now removed from your computer. The program files and desktop icon are also deleted automatically.

9. Software Interface

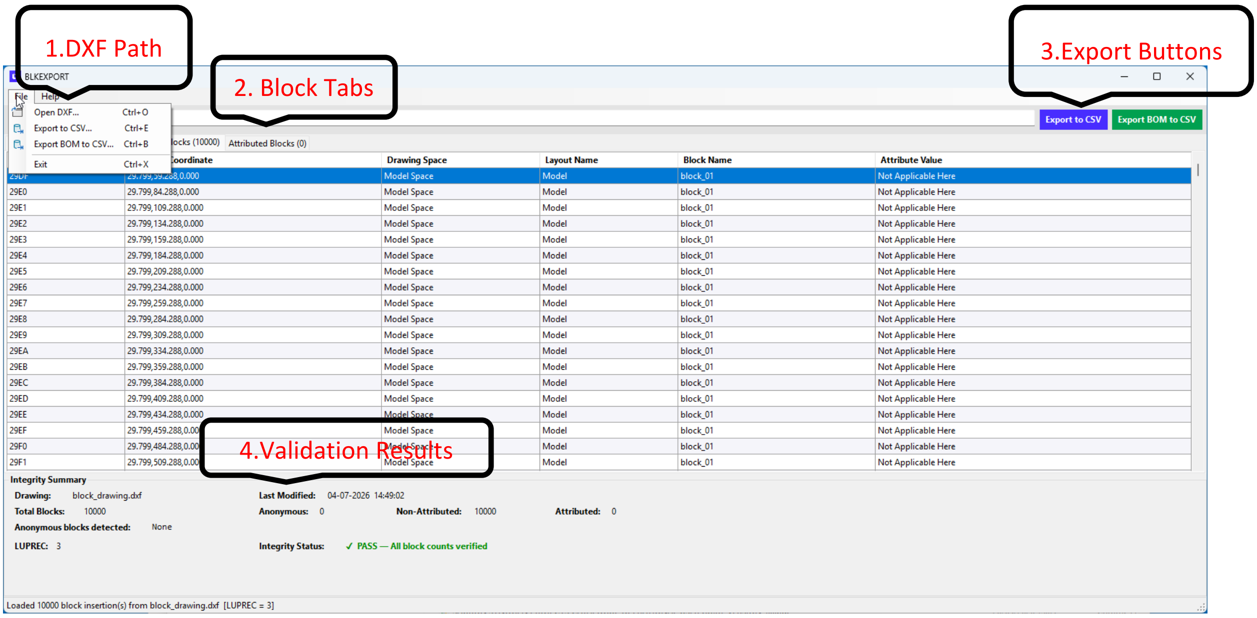

Figure 1 ***Main window after loading a DXF. Note the 3 block tabs for filtering and color-coded export buttons — purple for full schedule, green for BOM. Check the Integrity Summary bottom-right to confirm PASS before exporting***

10. Menu Commands

File Menu:

• Open DXF (Ctrl+O) – Open and process a DXF drawing.
• Export to CSV (Ctrl+E) – Export full block schedule data purple button.
• Export BOM to CSV (Ctrl+B) – Export a summarised BOM (Bill of Materials) green button.
• Exit (Ctrl+X) – Close the application.

Help Menu:

• About – Displays application version, registration information and system information.

11. Loading a DXF File

1. Select File > Open DXF or press Ctrl+O.
2. Browse to the required DXF file.
3. The application processes the drawing and populates the block tables.
4. Review the Integrity Summary to confirm successful processing.

12. Block Categories

Anonymous Blocks – Displays anonymous block references detected in the drawing.
Non-Attributed Blocks – Lists blocks without attributes.
Attributed Blocks – Lists blocks containing attribute values.

13. Data Columns

Typical columns include:

• Handle – Unique entity handle.
• Insertion Coordinate – Block insertion point.
• Drawing Space – Model Space or Paper Space.
• Layout Name – Associated layout.
• Block Name – Name of the inserted block.
• Attribute Value – Attribute content when applicable.

14. Exporting Data

Export to CSV:

1. Load a DXF file.
2. Click Export to CSV purple button or press Ctrl+E.
3. Choose a save location.
4. Open the generated full block schedule CSV in Excel or another spreadsheet application.

Export BOM to CSV:

1. Click Export BOM to CSV green button or press Ctrl+B.
2. Select a destination folder.
3. Open the generated summarised BOM (Bill of Materials) CSV in Excel or another spreadsheet application.

15. Integrity Summary

The Integrity Summary section provides:

• Drawing name (dxf) and modification date.
• Total block count.
• Anonymous, attributed, and non-attributed block counts.
• LUPREC value used for coordinate precision.
• PASS/FAIL status confirming that detected block counts have been verified.

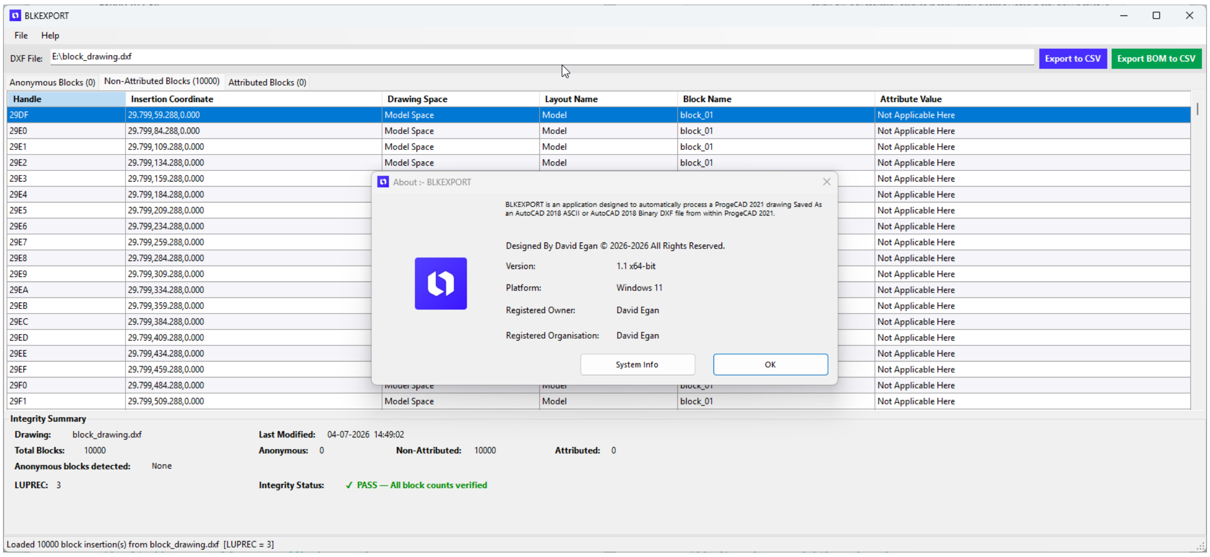

Figure 2

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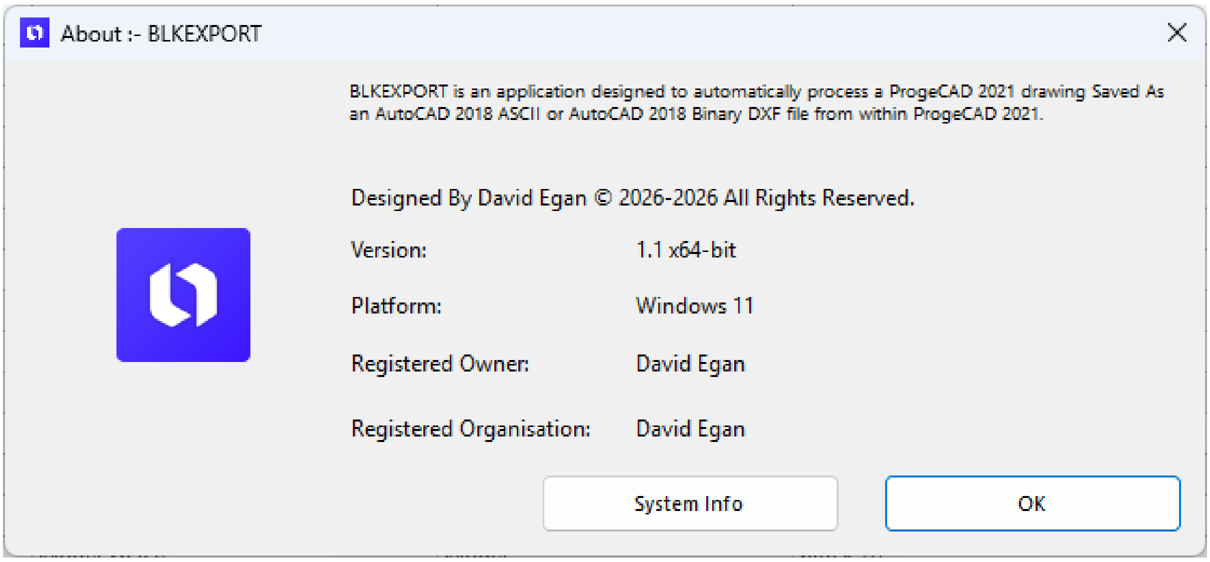

Figure 3 The About dialog displays version information, platform support, registration details, and access to system information

16. Software Design Philosophy

No AI. No fluff. Just engineering.

BLKEXPORT is built like the structures I trust — with reinforced concrete and steel.

What that means:

- **Predictable**: Does exactly what you expect, every time you run it.
- **Dependable**: No hidden calls home. No random updates breaking your workflow.
- **Built to last**: Engineered for real CAD work, not demos or trends.

I build software the way it was 20 years ago: solve the actual problem, then get out of your way.

Like a good bridge, it should work on day 1 and day 1001. Same result. No surprises.

That's the standard I hold myself to.
That's BLKEXPORT.

David Egan
Sole Software Developer & Systems Architect

17. Troubleshooting

No data displayed:

• Verify the selected file is a valid DXF.

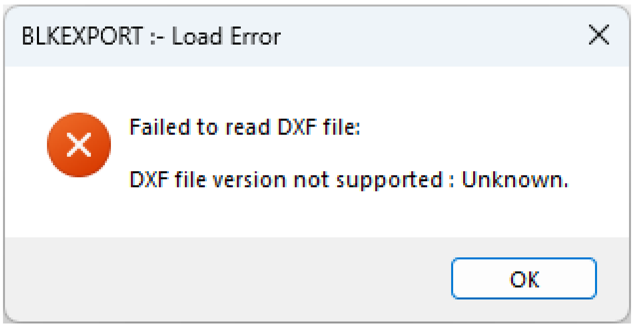

Figure 4 Invalid DXF file selected

• Ensure the drawing contains block insertions.

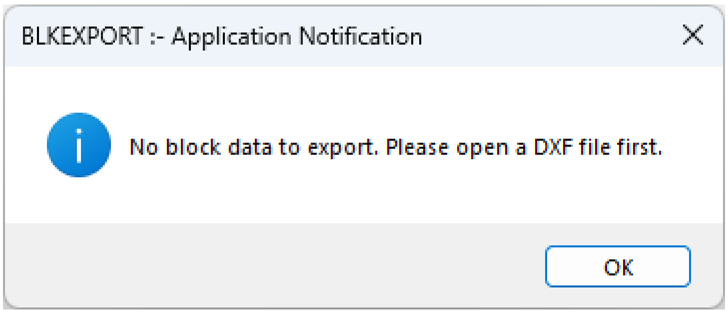

Figure 5 No blocks in the drawing to export or there is no drawing selected

Export unavailable:

• Load and process a DXF file first.
• Verify you have write permission to the target folder.

Integrity check failure:

• Re-save the drawing as a supported DXF format and reload it.

18. Licensing & Registration

Licence type: BLKEXPORT v1.1 is licensed as perpetual, single-user software.

Activation: No internet connection or activation required. The software runs fully offline.

Registration: Registration details are displayed in Help > About. For commercial licences, the registered company name appears here. Unregistered copies display “Unregistered”.

Trial limits: There is no time-limited trial. Unregistered copies are fully functional.

No calls home: BLKEXPORT does not contact external servers for licence checks, updates, or telemetry.

If you require a commercial licence or volume licensing, contact: licensing@blkexport.com

19. Updates & Support

Software updates: BLKEXPORT does not auto-update. This is intentional — no random updates breaking your workflow. New versions are announced at www.blkexport.com. To update, download the new installer and run it. It will replace the previous version.

Support policy: BLKEXPORT is provided as-is. For bugs that prevent core functionality — loading a valid DXF or exporting CSV — report them to support@blkexport.com with:

1. BLKEXPORT version from Help > About
2. Windows version
3. The DXF file, if possible
4. Screenshot of the Integrity Summary

Response is on a best-effort basis. Feature requests are logged but not guaranteed.

No phone support: Email only. This keeps the software cost down and your workflow uninterrupted.

20. Known Issues in v1.1

No known issues affecting data integrity as of June 2026.

Minor behaviour notes:

Large file loading: The UI may appear frozen during DXF processing. This is expected. See "DXF load performance" in Section 5.

Network paths: Opening DXF files from network drives works, but local SSD is recommended for files >10MB to avoid timeouts.

High-DPI displays: On some 4K monitors with >150% scaling, button text may clip. Functionality is unaffected.

If you encounter an issue not listed here, see Section 17 Troubleshooting first, then Section 21 Updates & Support.

21. Keyboard Shortcuts

Ctrl+O – Open DXF file
Ctrl+E – Export to CSV in purple button
Ctrl+B – Export BOM to CSV in green button
Ctrl+X – Exit

22. Glossary

Anonymous Blocks: AutoCAD-generated blocks with names like *U1, *D123. Typically created by dimensions, hatches, or dynamic blocks. BLKEXPORT lists them separately as they cannot be renamed meaningfully.

Attributed Blocks: Blocks that contain attribute definitions for storing text data, e.g. part numbers or descriptions.

BOM: Bill of Materials. A summarised list of block names and quantities.

DXF: Drawing Exchange Format. An ASCII or binary file format for CAD data interchange. BLKEXPORT supports DXF only.

Entity Handle: A unique hexadecimal identifier assigned to every object in an AutoCAD drawing.

LUPREC: AutoCAD system variable controlling the decimal precision displayed for linear units. BLKEXPORT uses this value when rounding block insertion coordinates in exports.

Non-Attributed Blocks: Blocks without attribute definitions. Only the block name and insertion data are exported.

Paper Space / Model Space: AutoCAD drawing contexts. BLKEXPORT reports which space each block insertion belongs to.

23. Version History

v1.1 – June 2026

1. Added “Tested Limits” and reference test machine specification to Section 5.
2. Clarified DWG is not supported in Section 6.
3. Added Section 24: The importance of meaningful block names.
4. UI text updated to UK spelling: “summarised”, “categorises”.

v1.0 – March 2026
Initial release.

24. The Importance of Meaningful Block Names

Using meaningful block names makes the Summarised BOM and Full Block Schedule CSV exports much easier to read and understand.

If your CSV exports contain obscure or generic block names, rename the blocks in your CAD software before exporting. You can use the Rename Blocks command in AutoCAD or progeCAD 2021 to assign descriptive, meaningful names to your blocks.

After renaming the blocks, save the DXF file and close it in your CAD application. The DXF file must be closed before opening it in the BLKEXPORT software.

Once the DXF file has been processed in BLKEXPORT, export the Summarised BOM or Full Block Schedule CSV again. The exported CSV files will now display your newly assigned block names, making them easier to interpret.

25. Preparing Symbols Sheets for Use with BLKEXPORT

If you plan to incorporate BLKEXPORT software into your drawing workflow, you may need to modify your symbols sheet drawings. This typically involves renaming the blocks within the sheet to give each symbol a clear, meaningful name.
Completing this exercise up front is highly beneficial — it eliminates the need to rename blocks for all future drawings. For existing drawings, however, block renaming will likely still be required.

Best Practice from Experience

When I worked as a CAD Technician, I would always copy only the required symbols from the latest symbols sheet into a new drawing. This ensured I was consistently using the most current, approved symbols and saved significant time by avoiding rework or renaming blocks after the drawing was complete.

There is a strong emphasis on the person producing the drawing to exercise due diligence during drawing preparation. Specifically, symbols must be created as blocks where appropriate, rather than drawn as raw geometry.

Critical Requirement

BLKEXPORT will only export objects that are defined as blocks within the drawing. If symbols are not created as blocks, BLKEXPORT cannot export them.

26. Comparing the Summarised BOM with the Source Drawing

Once the drawing is complete and the summarised Bill of Materials (BOM) has been generated, it is good practice to verify the BOM against the symbols in the actual drawing. Print both the drawing and the summarised BOM and compare them.

Traditionally, the BOM is produced manually by printing the drawing, then using a highlighter to count each symbol by hand and record the quantities.

BLKEXPORT software eliminates this manual process, generating the BOM faster and with greater accuracy. However, the accuracy of the summarised BOM depends entirely on the use of blocks in the drawing. There is a strong emphasis on the person producing the drawing to exercise due diligence during drawing preparation — specifically, ensuring that symbols are created as blocks where appropriate, rather than drawn as raw geometry.

BLKEXPORT will only export objects that are defined as blocks in the drawing. If symbols are not created as blocks, BLKEXPORT cannot export them.

27. Symbol Placement and BOM Accuracy

As a rule, symbols belong in model space. You'll still see them in paper space sometimes, but good drawing practice is to keep all symbols in model space.

To get a complete inventory of every symbol in your drawing, use BLKEXPORT and export a full block schedule to CSV. That report lists all symbols present in the file.

If you need an accurate summarised BOM, remove any symbols that are sitting in paper space before you export. Paper space symbols are often part of a drawing legend or title block. Including them can skew the BOM counts, so it's best practice to delete or relocate them to model space first.

Why it matters:

- **Model space =** real-world drawing elements that make up the design
- **Paper space =** sheet layout, legends, notes, title blocks
- Symbols in paper space don't represent actual installed components, so they inflate BOM quantities if left in

Quick workflow:

1. Export full block schedule with BLKEXPORT → see all symbols
2. Check for paper space symbols, especially in legends
3. Remove or move them to model space if needed
4. Export the summarised BOM for accurate counts

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